

Garden Presentations of the CLS Valise

Exact scan of the census G-matrix representatives, and closure of the equivalence question

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This is the companion to the CLS valise G-matrix census précis (docs/cls-gmatrix-census-precis-for-gates.tex). The census found 123,984,864 matrices $G \in \{-1, 0, 1\}^{12 \times 12}$ with $G^2 = A$ and stored 4,536 class representatives (4,077 distinct matrices). The natural follow-up, prompted by the holoraumy program of [1], is: *which* stored G conjugate all four CLS color matrices L_I into signed-permutation (Garden) form? Exact answer over the stored record: **exactly 9 of the 4,077**, all first stored in the two highest-mass census items. These nine qualifying matrices produce three Garden-presentation quadruples, and entrywise-verified signed node relabelings carry every one of them to the CLS presentation with the color index untouched. Consequences: exactly **one signed-node-relabeling equivalence class** among the stored Garden presentations; their holoraumy tuples are conjugate, rather than entrywise identical, and carry the same invariant content. The nine G -matrices are integral, non-monomial basis changes connecting the CLS monomial presentation to signed node relabelings of itself. An internal preliminary extrapolation of 33,696 qualifying matrices was rejected before delivery because the required taxonomy-constancy hypothesis is false.

The question

The census précis enumerated all $G \in \{-1, 0, 1\}^{12 \times 12}$ satisfying $G^2 = A$ with $A = (L_1 + L_2 + L_3 + L_4) L_1^{-1}$ for the CLS color matrices of [1, Appendix C]. The CLS L_I are signed permutations satisfying the Garden algebra entrywise,

$$L_I L_J^T + L_J L_I^T = 2 \delta_{IJ} I_{12},$$

with multiplicative companions $L_I^2 = -I$ for colors 2, 3, 4 (color 1 is the identity) and anticommutation among colors 2, 3, 4.

Conjugation $C_I = G L_I G^{-1}$ preserves the multiplicative relations for any invertible G but destroys monomiality in general. A *Garden presentation* here means the ordered quadruple $C = (C_1, C_2, C_3, C_4)$ when all four C_I are integer signed-permutation matrices. A census matrix G *qualifies* when it produces such a quadruple. A qualifying G turns the valise back into a monomial adinkra presentation in a different basis, which is the entry point for comparing holoraumy data across bases. The scan asks, exactly, how often that happens among the stored census representatives and what the resulting presentations are up to equivalence.

Exact scan of the stored representatives

Method (scripts/lever_a/garden_presentation_scan.py, exact Fraction arithmetic, no floating point anywhere, one core, about 50 seconds):

- The four L_I are transcribed in signed-address form from src/four_color/cls.rs (Appendix C of [1]), with $L_1 = I_{12}$.
- Sanity anchors, all verified entrywise before any scan: the four L_I are signed permutations; the Garden algebra above holds; $A = (L_1 + L_2 + L_3 + L_4) L_1^{-1}$ matches the census artifact results/four_color_cls_gmatrix.json entrywise, tying the color matrices to the census target; $L_2^2 = L_3^2 = L_4^2 = -I$ with $L_1^2 = +I$; colors 2, 3, 4 pairwise anticommute.
- Input: all 138 shard files across the three canonical census run directories; every stored class representative: 4,536 records, 4,077 distinct matrices, 1,076 taxonomy keys (nnz, support, ranks).
- Classification by exact rational inverse and matrix product: bucket 1 if all four C_I are integer signed permutations, bucket 2 if all integer but at least one not signed-permutation, bucket 3 if any entry is non-integer.

Classification over the 4,077 distinct stored matrices	Count
Qualifying G : all four colors signed-permutation	9
Integer, only partially signed-permutation	228
At least one non-integer conjugated color	3,840

Provenance of the nine: 7 are first stored in the shard of census item 52 and 2 in the shard of item 204, the two items contributing the largest census mass. All nine also satisfy the Garden algebra entrywise on their conjugated colors, but this is a corollary, not an extra filter: signed-permutation conjugates are orthogonal. For $I = 2, 3, 4$, the similarity-preserved relation $C_I^2 = -I$ therefore gives $C_I^T = C_I^{-1} = -C_I$. Anticommutation among colors 2, 3, 4 then gives $C_I C_J^T + C_J C_I^T = 0$ for distinct I, J in that range. For a pair involving color 1, $C_1 = I$ and $C_J^T = -C_J$, so $C_1 C_J^T + C_J C_1^T = C_J^T + C_J = 0$. The diagonal relations follow directly from orthogonality.

Why no population count follows

Two traps close off any extrapolation from the nine to the full 123,984,864.

Taxonomy non-constancy. The property is not constant on the census taxonomy classes. Counts of keys whose stored representatives disagree, by explicit definition of the classification:

Definition of the classification	Mixed keys
Coarse bucket (3-way)	34
Full four-color signed-permutation flags	50
Combined (bucket, signed-permutation flags)	55
Number of integer colors	182
Full four-color integer flags	188

Every definition is far above zero, so a taxonomy key does not determine the conjugation property. The internal preliminary estimate that 33,696 matrices qualify (extrapolated by assuming constancy on classes) is therefore unsupported and was rejected before delivery. Nothing in the shipped census précis ever carried that number; this document records the validated position.

Similarity tautologies. Multiplicative relations among the C_I are preserved by any similarity and carry no information beyond $C_I^2 = GL_I^2 G^{-1}$. Thus $C_1^2 = +I$ and $C_2^2 = C_3^2 = C_4^2 = -I$ are forced and are not evidence of an additional property. Only the non-similarity-invariant structure (integrality, monomiality of the conjugated colors, and the transpose algebra, which for bucket 1 follows from monomiality as shown above) has content.

The nine collapse to three quadruples

The nine bucket-1 matrices produce only *three* distinct conjugated quadruples $T^{(1)}, T^{(2)}, T^{(3)}$: seven of the nine land on $T^{(1)}$ and one each on $T^{(2)}$ and $T^{(3)}$. The collapse mechanism is verified directly: within each group, $G_j^{-1} G_i$ commutes with all four L_I (commutant collisions), so the different G 's differ by elements of the commutant of the color algebra rather than by genuine presentation changes.

One class: exact signed-monomial equivalence closure

For ordered presentations $C^{(a)}$ and $C^{(b)}$, the tested adinkra equivalence relation asks for a signed node map S , a color permutation σ , and optional color sign rescalings $\epsilon_I \in \{\pm 1\}$ such that

$$S C_I^{(a)} S^{-1} = \epsilon_I C_{\sigma(I)}^{(b)}.$$

Signs carried by the signed monomial matrix S are node signs; they are distinct from the optional color signs ϵ_I . The executed result has $\sigma = \text{id}$ and $\epsilon = (1, 1, 1, 1)$ for every displayed witness:

All four quadruples (CLS itself, $T^{(1)}$, $T^{(2)}$, $T^{(3)}$) form **one equivalence class under signed node relabeling alone**: σ is the identity, and no color permutation or color sign rescaling is needed. The signed node intertwiners are verified entrywise ($SL_I S^{-1} = T_I$ for all four colors).

In signed-address form (row i carries its single nonzero, $\text{sign}(a_i)$ in column $|a_i|$):

Map	Intertwiner S (address form)
$\text{CLS} \rightarrow T^{(1)}$	(1, 4, 2, 3, 5, -7, 8, -6, 9, -11, -12, 10)
$\text{CLS} \rightarrow T^{(2)}$	(5, -7, 6, -8, 1, 4, -3, -2, 9, -11, -12, 10)
$\text{CLS} \rightarrow T^{(3)}$	(5, -8, -7, 6, 9, 11, 12, 10, 1, -2, 4, -3)

The conclusion that the four displayed presentations form one class is constructive: the three witnesses above already prove it. The accompanying search also exhausts the stated signed-monomial equivalence relation. Each quadruple is block-diagonal 4+4+4, and each 4-node block is a connected component of the union of its colored node graph. A signed monomial intertwiner preserves this colored adjacency and therefore maps each complete connected block onto a complete connected block. Its support consequently contains exactly one 4×4 signed-permutation block in each block row and block column. The search enumerates all $3!$ block assignments, all $4! 2^4 = 384$ signed permutations inside each matched block, all allowed global color permutations, and all color sign choices. Block connectivity, the resulting plans, and every displayed witness are checked entrywise by `scripts/lever_a/garden_target_equivalence_check.py`.

The block plans make the collapse structural. Writing each T as a triple of color permutations of the original CLS blocks: $T^{(1)}$ uses $(234)/(243)/(243)$, $T^{(2)}$ uses $(23)/(24)/(243)$, and $T^{(3)}$ uses $(24)/(243)/(34)$. Within a block family the inducible pure color permutations are exactly the even ones (the identity and the two 3-cycles); cross-family block conjugators induce exactly the transpositions. Every plan is realizable by signed node relabeling, which is why everything collapses to one class.

Holoraumy consequence, and a cross-gadget artifact

Because the nine qualifying matrices produce presentations related to CLS by signed node relabelings, their holoraumy tuples are conjugate by the same relabelings. They are not generally entrywise identical in the displayed bases. There is exactly one holoraumy equivalence class among the stored Garden presentations, so no new invariant content is present.

One recorded misreading is worth stating precisely, because it is a trap for any future basis comparison. The holoraumy gadget (`src/holoraumy.rs`) was evaluated over the four quadruples, summing traces of products of the $\tilde{V}$ matrices of one presentation against those of another. The resulting cross-matrix,

	CLS	$T^{(1)}$	$T^{(2)}$	$T^{(3)}$
CLS	3	0	0	0
$T^{(1)}$	0	3	1	1
$T^{(2)}$	0	1	3	0
$T^{(3)}$	0	1	0	3

was briefly read as “CLS is holoraumy-orthogonal to its conjugates.” That reading refutes itself: every target is a conjugate of CLS by construction, so an isomorphism-invariant comparison must return the self-value 3 on

every entry. The observed 0s and 1s prove the cross-gadget is a functional of a pair of *presentations*, not of a pair of isomorphism classes: the trace of a product of a matrix from one basis against a matrix from another basis moves under independent conjugation of the two factors. Direct certificate: the witness relabeling that carries $T^{(2)}$ onto $T^{(1)}$ moves the $(T^{(1)}, T^{(2)})$ entry from 1 to the self-value 3. This basis dependence applies to every pair of presentations, whether their underlying representations are equivalent or inequivalent. The raw cross-gadget is therefore a basis-alignment functional, not by itself a representation-level comparison. Such a comparison requires a common basis convention, an explicit alignment prescription, optimization over the allowed relabeling orbit, or a separately proved invariant construction.

What the nine G -matrices actually are, then: integral, non-monomial basis changes connecting the CLS monomial presentation to signed node relabelings of itself. The narrow exact fact that survives is the scan headline: nine of the 4,077 stored representatives conjugate into signed-permutation form, and that presentation is unique up to signed node relabeling.

What remains open

Whether *additional* matrices in the nine taxonomy classes (beyond the stored representatives) also conjugate into Garden form is unknown. Testing that requires materializing class members, which needs the census solution-dump hook and a targeted rerun of the two affected census items, items 52 and 204, while materializing members of the nine affected taxonomy classes. It does not require a new census. Until that runs, no population count of any size should be attached to the property.

Artifacts and reproduction

The computational record is pinned to repository <https://github.com/plp3dream/adinkra-codespace> at commit 69fe57ba230b0dbd8cbc677b0c1b00b08c5b94b6. That commit fixes the numerical scan, the three target quadruples, the signed-node witnesses, and the cross-gadget rebasing certificate. This précis and its compiled PDF are versioned beside the supporting artifacts under docs/.

- results/cls-garden-presentation-scan-20260822.md: full writeup with correction log
- scripts/lever_a/garden_presentation_scan.py: the exact scan; python3 it, runs in about 50 seconds on one core, reproduces 9/228/3,840 and all bookkeeping counts (4,536 records, 4,077 distinct, 1,076 keys, 712 multi-representative keys, mixed-key counts 34/50/55/182/188)
- Equivalence check, directory scripts/lever_a/: garden_target_equivalence_check.py. It performs the commutant-collapse checks, connectivity gates, intertwiner verification, and complete signed-monomial block search; all checks pass
- src/four_color/cls.rs: the CLS L -matrices in signed-address form
- Companion: docs/cls-gmatrix-census-precis-for-gates.tex (the census itself)
- This document: docs/cls-garden-presentation-equivalence-precis-for-gates.tex and its generated PDF

References

- [1] S. J. Gates Jr. and Y. Lee, “A Précis: Minimal Four Color Holoraumy and Wolfram’s ‘New Kind of Science’ Paradigm,” preprint (2024), [arXiv:2408.09342](https://arxiv.org/abs/2408.09342) [hep-th].